

DFD-NAS: General deepfake detection via efficient neural architecture search[☆]

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ARTICLE INFO

Communicated by G. Yang

Keywords:

Deepfake detection
Face forgery detection
Video forensics
Neural Architecture Search (NAS)

ABSTRACT

As the saying goes, “seeing is believing”. However, with the development of digital face editing tools, we can no longer trust what we can see. Though face forgery detection has made promising progress, most current methods are manually designed by human experts, which is labor-intensive. In this paper, we develop an end-to-end framework based on neural architecture search (NAS) for deepfake detection, which can automatically design network architectures without human intervention. First, a forgery-oriented search space is created to choose appropriate operations for this task, which facilitates the search process in finding the most valuable gradient information for face forgery detection. Second, inspired by the fact that the gap between training error and test error is a good indicator of generalization ability for a classification task, we propose a novel performance estimation metric. This metric encourages the error in the performance estimation phase to be close to the error in the search phase, which guides the search process to select more general models. The cross-dataset search is also considered to develop more general architectures. Eventually, we design a cell cascaded pyramid network (C2PN) for final detection, which aggregates multiscale features for performance improvements. Compared with state-of-the-art networks artificially designed, our method achieves competitive performance in both in-dataset and cross-dataset scenarios.

1. Introduction

The emergence of face forgery technology has caused serious concerns in the whole society [1–5]. First, using photos of political figures, stars, and other people may incur legal issues, such as reputations, portrait rights, intellectual properties, and so on. Second, videos of important persons are highly valuable intelligence. The abuse of counterfeit technology can lead to the spread of false information, even cause diplomatic crises. In addition, privacy disclosure may further affect asset security and personal safety. The fraud of such vivid deepfakes will cause a number of legal, social, and ethical problems.

To fight against the abuse of facial manipulation technologies, many pioneers have made significant progress [6–8]. However, the problem is far from completely solved. On one hand, though decent performance has been achieved when training and tests are implemented in the same dataset, the generalization ability in the cross-dataset scenario is still challenging. There is an urgent demand for more general architectures. On the other hand, most of these methods are hand-crafted by human

experts, which is labor-consuming. The emergence of neural architecture search (NAS) makes it possible to automatically design neural networks for deepfake detection.

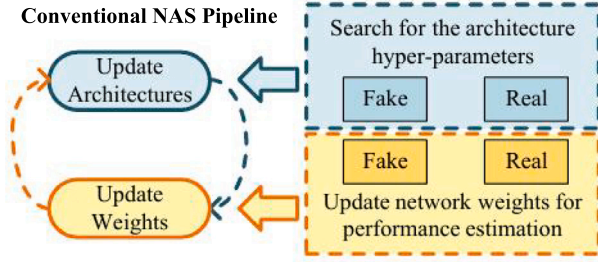
In this paper, we mainly focus on two challenges of face forgery detection. The first one is the limitation of the performance in cross-dataset settings. Existing methods often suffer from performance degradation when the training set is different from the test set. The forgery tools and video contents can vary significantly between two datasets. Forgery tools have diverse principles and processes, leaving distinct clues in the video. In addition, video content also affects the practicability of some existing techniques. Due to the difference in skin colors and facial features of persons, the performance of classifiers may decrease to a certain degree [6,9,10].

There are some prior arts for general face forgery detection. The previous approaches mainly aim to extract more general features [11–13]. However, feature extraction is only a part of a neural network. How to design a general detection framework as a whole has attracted relatively much less attention. Additionally, till now, most networks for

[☆] This work was supported in part by the National Natural Science Foundation of China (62401295), Tianjin Natural Science Foundation, China (22JC-QNJ01650), Fundamental Research Funds for the Central Universities of Nankai University (63201192, 63211116), Wukong Investment - Entropy Control Network Fund (078-G02J0455).

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Our contributions:

- (1) create a forgery-oriented search space,
- (2) introduce a novel performance estimation metric.

Fig. 1. Conventional NAS pipeline mainly consists of two parts, searching for the architecture hyper-parameter and updating network weights for performance estimation. (1) For the first part, we observe that the manipulation traces are too fragile to be discovered by ordinary layers in the neural networks. The gradient information plays a vital role in deepfake detection. The value of θ in central difference convolution (CDC) controls the proportion of gradient information. Thus, we gradually change the value of θ in the search space. (2) For the second part, we notice that some NAS methods are prone to overfitting, while generalization capability is of importance for facial manipulation detection. Therefore, we introduce a novel performance estimation indicator, encouraging the error in the performance estimation phase to be close to the error in the search phase.

face forgery detection are manually designed, which requires a lot of experience from experts.

The second challenge is the limitation of manually designed networks. Artificially designed neural networks have achieved good performance, while they require rich professional knowledge and massive repeated experiments. As the number of network layers increases, more hyper-parameters need to be manually adjusted, which consumes labor, time, and computing resources. This situation limits the further development of this field to some extent. Fortunately, the emergence of NAS technology provides a possibility to solve deepfake detection from a new perspective.

Though numerous NAS algorithms have been proposed, the inherent difference between face forgery detection and other computer vision tasks makes it difficult to adopt other NAS methods directly. First, the manipulation traces are too fragile to be discovered by ordinary layers in the neural networks. Second, some NAS methods are prone to overfitting, while generalization capability is of importance for facial manipulation detection. Third, increasing the number of operations in the candidate search space causes an inefficient search process.

The main contributions of this paper are illustrated in Fig. 1. In summary, our contributions are as follows:

- We propose a novel end-to-end NAS framework to automatically design a network architecture for face forgery detection. To better fit this task, we create a forgery-oriented search space by gradually select appropriate gradient information. To the best of our knowledge, it is the first NAS-based method with specialized search space for deepfake forensics.
- Additionally, inspired by the fact that the distance between training error and test error is a typical indicator of generalization ability for a classification task, we also introduce a novel performance estimation metric. The proposed metric can select more general architectures by encouraging the error in the performance estimation phase to be close to the error in the search phase.
- Based on the cells obtained by NAS, we further develop a cell cascaded pyramid network (C2PN) for final face forgery detection, which incorporates multiscale features for performance improvements. Extensive experiments and visualizations demonstrate the effectiveness of the proposed method on four benchmark datasets.

The rest of the paper is structured as follows. In Section 2, we introduce the related work. Then DeepFake Detection (DFD) framework is presented in Section 3. The experimental result and analysis will be detailed in Sections Section 4. The conclusion of the paper is summarized in Section 5.

2. Related works

2.1. Face forgery detection

Some prior arts are proposed for face forgery detection [2–5,14]. In this subsection, we summarize the previous works by the architectures they choose. In the early years, researchers explore hand-crafted features for face forgery detection. Qi et al. [15] propose DeepRhythm to classify deepfakes by monitoring the heartbeat rhythms. Agarwal et al. [16] detect face forgery by modeling facial expressions and movements that character the speaking patterns. Yang et al. [17] observe that deepfakes are generated by splicing synthesized face regions, and present a forensics pipeline by 3D head pose estimation.

The performance of the above methods based on hand-crafted features is quickly surpassed by the forensic framework designed by deep learning. Nirkin et al. [18] design a method for detecting fake videos using a single frame. This method includes a face recognition network and a context recognition network for facial features. Yang et al. [19] adopt central difference convolution and atrous spatial pyramid pooling (ASPP) to propose a multi-scale texture difference model for face forgery detection. Yu et al. [13] introduce a specific forgery feature extractor and a common forgery feature extractor for learning universal forensic features. Wang et al. [20] use a cross-modality knowledge distillation method for near-infrared face forgery authentication. Nevertheless, as the structure of neural networks becomes more and more complex, the hyperparameters that need to be adjusted increase geometrically. Designing networks manually for deepfake detection becomes labor-consuming due to repetitive experiments.

With the popularity of Transformer architecture, more and more detection methods based on this technique appeared. Miao et al. [21] attempt to exploit this problem from both spatial and frequency domains. They design a high-frequency fine-grained Transformer network, which consists of central difference attention and high-frequency wavelet samplers. Zhao et al. [22] employ an interpretable spatial-temporal video Transformer to capture spatial artifacts and temporal inconsistency for robust deepfake detection. This framework is composed of a decomposed spatial-temporal self-attention and a self-subtract mechanism. Though Transformer-based methods bring performance improvements, the architectures designed by human experts require substantial experience and repeated experiments.

Adopting NAS framework for face forgery detection receives relatively less research attention. Liu et al. [23] develop an automated deepfake detection framework using common convolutional layers as the search space. However, ordinary convolution is not specially designed to extract forgery traces. There is still room for performance improvement if the search space is appropriately selected.

2.2. Neural architecture search

Neural architecture search has proposed for automatically design neural networks [24]. The NASNet [25] is the first algorithm to use modular search, which greatly reduces the computation time due to normal cells and reduction cells. The ENAS [26] samples subgraphs from a large hypergraph, and the operations in the subgraph inherit the weights of the hypergraph. This approach facilitates the sharing of weights between different structures and greatly reduces the training time of neural network structures in performance prediction. The ProxylessNAS [27] searches based on MobileNetV2, changing the number of convolutional layers, convolutional kernel sizes, and the number of filters in its internal MBConv unit. All MBConv units have no shared

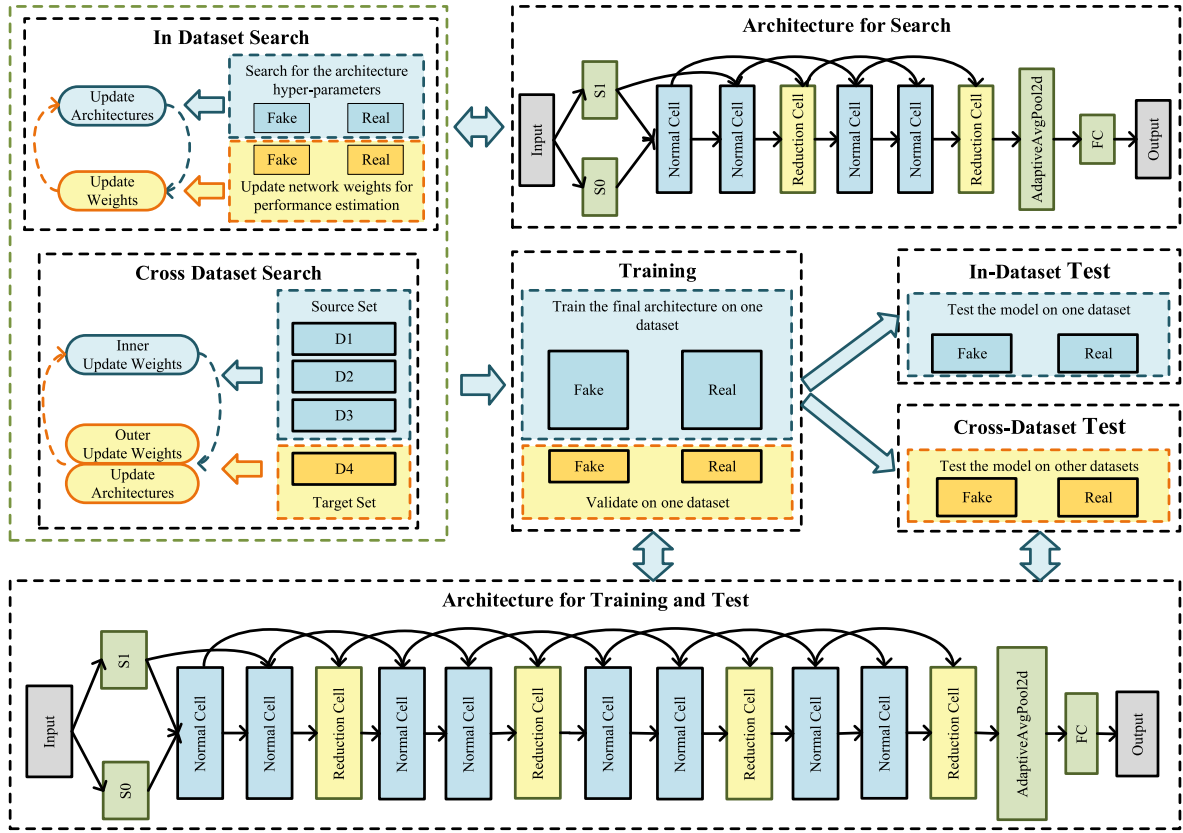


Fig. 2. The framework of the proposed method. There are three main steps in our approach: search, training, and test. In the search step, differentiable architecture search updates the architecture hyper-parameter and the network weights alternately. Due to the limitation of GPU memory, we adopt a lightweight network in this step. In training and test steps, we employ a deeper network to preserve valuable multiscale information for face forgery detection.

structure and therefore have more flexibility. The MnasNet [28] formulates the search process as a multi-objective optimization problem, considering both model accuracy and inference latency. The differentiable architecture search (DARTS) [29] defines the search space as a continuous space. Each searched cell is derivable and can be optimized by stochastic gradient descent.

Ge et al. [30,31] utilize DARTS and its variant for anti-spoofing and speech authenticity identification. Wu et al. [32] propose an approach based on NAS for image inpainting detection. However, they only use NAS for designing an extraction block. The forgery clues are still discovered by manually designed layers, which is not an end-to-end NAS framework.

Furthermore, due to the intrinsic discrepancy between deepfake detection and other computer vision tasks, directly applying other NAS frameworks has not achieved satisfactory results in face forgery detection [23,33]. There is still room for performance increase if the framework is specifically designed for deepfake detection. How to analyze fake videos from a NAS-based perspective is still an open problem.

3. The proposed method

In this section, we first revisit the differentiable architecture search process. Then, we describe the search space, cell topology, performance estimation strategy, search strategy, and cross-dataset search, respectively. After that, we suggest the C2PN for final detection. Eventually, we also discuss the novelty of the proposed method. The whole framework is illustrated in Fig. 2.

3.1. Revisit differentiable architecture search

The process of neural architecture search mainly includes three parts: search space, search strategy and performance estimation. First, the search space defines all the candidate neural network operations or components. Then, the corresponding search strategy selects a certain neural network structure from the space. This intermediate network is evaluated by several performance indicators. Finally, the evaluation results are fed back to the search strategy until the optimal architectures are obtained.

Differentiable architecture search (DARTS) regards operation selection as a weighted combination of operator sets, so that the search space is relaxed to be continuous [29]. In this way, the hyperparameters can be calculated differentially by gradient descent. The basic component in the architecture is a cell, which can be considered as a directed acyclic graph (DAG) with N nodes. A topological node represents for a layer in the neural network. Usually, each cell is composed of two input nodes, one output node, and the remaining intermediate nodes. For the j th intermediate node, its output is a summation of all the forward results of preceding nodes,

$$\mathbf{x}_j = \sum_{i < j} f_{i,j}(\mathbf{x}_i), \quad 0 \leq i \leq j \leq N-1, \quad (1)$$

where $\mathbf{x}_i$ symbolizes the output feature map from the i th node, $f_{i,j}(\cdot)$ is the operation from the i th node to the j th node. In a cell, the directed edge between two nodes defines an operation, e.g., convolution, pooling, and activation function, selected from the search space $\mathcal{O}$. These

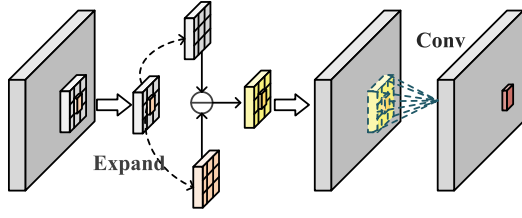


Fig. 3. The structure of central difference convolution.

operations can be formulated as,

$$f_{i,j}(\mathbf{x}_i) = \sum_{o \in \mathcal{O}} \frac{\exp\{\alpha_{i,j}^o\}}{\sum_{o' \in \mathcal{O}} \exp\{\alpha_{i,j}^{o'}\}} \cdot o(\mathbf{x}_i), \quad (2)$$

where $o(\cdot) \in \mathcal{O}$ indicates candidate operations, $\alpha_{i,j}^o$ is a trainable variable that determines the network architecture. Thus, the task of searching the network architecture is converted to optimizing all the $\alpha_{i,j}^o$ in the network. After the search process, every node is connected with two preceding nodes that have maximum $\alpha_{i,j}^o$ value. The final output architecture is constructed by several cells. These cells can be divided into two types, i.e., the normal cell that preserves the spatial resolution, and the reduction cell that decreases the resolution.

3.2. Search space

In this paper, the search space is mainly composed of central difference convolution (CDC). Fig. 3 shows the structure of CDC. Central difference operations have shown their effectiveness for face forgery detection. The features extracted with central difference can enhance the discriminability for forged images [19]. According to [19,34], the input is denoted as z , the output is represented by y , the process of CDC can be formulated as,

$$y(p_0) = \underbrace{\sum_{p_n \in \mathcal{R}} w(p_n) \cdot z(p_0 + p_n)}_{\text{vanilla convolution}} + \theta \cdot \underbrace{\left(-z(p_0) \cdot \sum_{p_n \in \mathcal{R}} w(p_n)\right)}_{\text{central difference term}}, \quad (3)$$

where $\mathcal{R}$ stands for local receptive field region, w is the weight, p_0 is the current location on feature maps, p_n is the other locations in $\mathcal{R}$, hyperparameter $\theta \in [0, 1]$ means the importance of gradient information.

In this type of operation space, we involve eight central difference operations with various parameter settings as shown in Fig. 4(a). ‘‘Sep’’ denotes ‘‘separable’’, while ‘‘Dil’’ represents for ‘‘dilated’’. For example, ‘‘SepCDC_3x3_0.5’’ is defined as a separable central difference convolutional layer, where the kernel size is 3×3 , and θ is 0.5.

When $\theta = 0$, central difference operations are converted to vanilla CNN. The vanilla CNN operations aim to discover some visual forgery artifacts, e.g., uncommon manipulated region boundaries. Concretely, since the multiscale information plays an important role in deepfake detection [19], we emphasize the skip connection and dilated convolution in the search space, which are denoted as ‘‘skip_connect’’ and ‘‘DilCDC_3x3’’, respectively.

In preliminary experiments, we also analyze the effectiveness of some noise extractors, e.g., Bayar layer [35] and SRM layer [36]. We find central difference operations perform better.

3.3. Cell topology

The topology structure of a cell is illustrated in Fig. 4(b), where ‘‘C’’ symbolizes concatenation. In our method, each cell contains seven nodes, including two input nodes, e.g., S0 and S1, four intermediate

Algorithm 1 The proposed method

Input: dataset D ; operation set $\mathcal{O}$

Output: cell topology

- 1: Split the dataset evenly into two parts, D_S for search, and D_{Eval} for evaluation;
- 2: Randomly sample a super-network and tune the network weights in the first 10 epochs;
- 3: **while** not converged **do**
- 4: Update architecture hyper-parameters on D_S ;
- 5: Update network weights on D_{Eval} ;
- 6: Update $\hat{E}_{i,j}^o$ and $\hat{E}_{i,j}^o$;
- 7: Calculate operation importance $H_{i,j}^o$ by Eq. (8);
- 8: Update each operation state S .
- 9: **if** epoch > 19 and epoch % 20 == 0 **then**
- 10: Prune operations according to their state S on each edge;
- 11: Update channel sampling rate $n = 2n$;
- 12: **end if**
- 13: **end while**
- 14: **return** cell topology

nodes, e.g., N0, N1, N2, and N3, and one output node. The edges indicate the operations searched from the forgery-oriented search space. At the beginning of the search, every node is linked with all predecessor nodes. At the end of the search, every node only keeps a connection with the two antecedent nodes with the highest scores. Finally, the search process produces two types of cells. The normal cells maintain the same resolution, while the reduction cells decrease the resolution of feature maps by half.

3.4. Performance estimation strategy

In previous gradient-based NAS methods, the significance of operations is ordered by architecture parameters $\alpha = \{\alpha_{i,j}^o, 0 \leq i \leq j \leq N-1\}$, which indicates the relative importance of an operation compared with other candidate operations. However, the operation with the highest value of α may not always bring the best performance [37]. Furthermore, DARTS tends to overfit [38], while the generalization ability on unseen datasets is significant to face forgery detection.

Therefore, we introduce additional conditions to select suitable operations,

$$E_{i,j}^o = \frac{\exp(1/\|\hat{E}_{i,j}^o - \hat{E}_{i,j}^o\|_{-\infty})}{\sum_{o' \in \mathcal{O}} \exp(1/\|\hat{E}_{i,j}^{o'} - \hat{E}_{i,j}^{o'}\|_{-\infty})}, \quad (4)$$

where $\hat{E}_{i,j}^o$ is the classification error in the search stage, $\hat{E}_{i,j}^o$ is the error in performance estimation process. Here, $\|\cdot\|_{-\infty}$ denotes the negative infinite norm, which returns the smallest absolute values of all elements. In this way, we encourage the error in performance estimation to be close to the error in search, improving the generalization capability of the network. The total estimation indicator is defined as,

$$I_{i,j}^o = \frac{\exp\{\alpha_{i,j}^o\}}{\sum_{o' \in \mathcal{O}} \exp\{\alpha_{i,j}^{o'}\}} + \lambda E_{i,j}^o, \quad (5)$$

where $\lambda = 0.15$ is the balance parameter.

3.5. Search strategy

To improve the efficiency of the search process, the feature maps are sampled by the proportion of $1/n$ in the channel dimension [39]. This process can be defined as,

$$f_{i,j}(\mathbf{x}_i; M_{i,j}) = \sum_{o \in \mathcal{O}} I_{i,j}^o \cdot o(M_{i,j} \odot \mathbf{x}_i) + (1 - M_{i,j}) \odot \mathbf{x}_i, \quad (6)$$

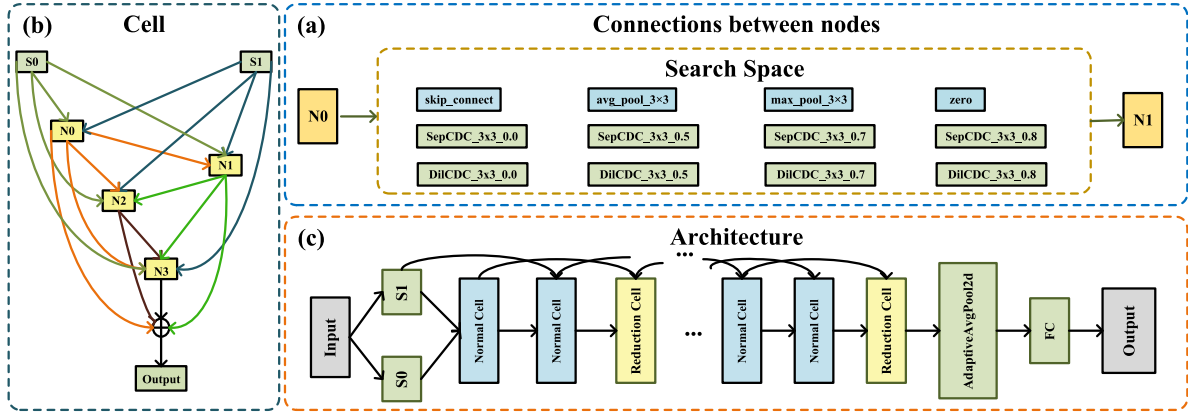


Fig. 4. The illustration of the NAS. (a) The connections between nodes in a cell. Each edge indicates an operation obtained from the search space. (b) The topological structure of a cell. (c) The architecture of the whole detection networks.

where $\odot$ represents the Hadamard product, $M_{i,j} \in \{0, 1\}$ is channel mask, 1 denotes this feature channel is taken, and 0 indicates this channel is discarded.

In the preliminary experiments, we find that the search process tends to select several types of operations, while neglecting other kinds of operations. To increase the diversity of the operations in the cell, we further take the edge parameter $\beta_{i,j}$ into consideration. The output of the j th intermediate node is defined as,

$$\mathbf{x}_j = \sum_{i < j} \frac{\exp\{\beta_{i,j}\}}{\sum_{i' < j} \exp\{\beta_{i',j}\}} \cdot f_{i,j}(\mathbf{x}_i). \quad (7)$$

The importance of the connections between edges is determined by $I_{i,j}^o$ and $\beta_{i,j}$. We define $H_{i,j}^o$ to describe this process,

$$H_{i,j}^o = \frac{\exp\{\beta_{i,j}\}}{\sum_{i' < j} \exp\{\beta_{i',j}\}} \cdot I_{i,j}^o. \quad (8)$$

The operation with the highest $H_{i,j}^o$ is adopted for final neural network architecture,

$$f_{i,j}^* = \arg \max_{o \in \mathcal{O}} H_{i,j}^o, \quad (9)$$

where $f_{i,j}^*$ denotes the final operation from the i th node to the j th node.

Moreover, to further accelerate the search procedure, we update the operation state $S \in \{\text{True}, \text{False}\}$ between two nodes according to $H_{i,j}^o$. True denotes this operation is retained for the following search, while False indicates this operation is discarded for the search in the next step. After every twenty epochs, we remove the two operations with the lowest scores, and increase the rate of channel sampling. The whole procedure is illustrated in Algorithm 1.

3.6. Cross-dataset search

A key challenge for deepfake detection is the performance degradation in cross-dataset scenarios. In this paper, we try to analyze this problem by automatically searching a general architecture. Inspired by [40], we adopt a cross-dataset search strategy. For simplicity, assuming we have K datasets $\mathcal{D} = \{D_1, D_2, \dots, D_K\}$, the randomly selected samples in $(K-1)$ datasets are defined as the source datasets $\mathcal{D}^s$, while the rest one is denoted as the target dataset $\mathcal{D}^t$. Then, the sample batch examples $I_i^s (i = 1, 2, \dots, K-1)$ in every dataset of $\mathcal{D}^s$, and I^t in $\mathcal{D}^t$ are employed for inner-updated network weights and architecture optimization stage, respectively. The process of cross-dataset search is described in Algorithm 2. Compared with [40], our work utilizes a more flexible search space to fully explore the potential of cross-dataset search strategy.

Algorithm 2 The cross-dataset search process

Input: dataset $\mathcal{D} = \{D_1, D_2, \dots, D_K\}$

Output: cell topology

- 1: Initialize weights and architecture;
- 2: **while** not converged **do**
- 3: Randomly select $(K-1)$ datasets in $\mathcal{D}$ as source datasets $\mathcal{D}^s$, and the rest one as target dataset $\mathcal{D}^t$;
- 4: Sample batch examples $I_i^s (i = 1, 2, \dots, K-1)$ in every dataset of $\mathcal{D}^s$, the examples I^t in $\mathcal{D}^t$;
- 5: **for** each I_i^s **do**
- 6: Update weights in the i -th dataset;
- 7: **end for**
- 8: Update weights using all learners' loss on the I^t ;
- 9: Update architecture on the I^t ;
- 10: **end while**
- 11: **return** cell topology

3.7. Cell cascaded pyramid network

After obtaining the cells, we connect them in series to detect face forgery videos. Since there are a large number of connections between nodes at the beginning of the search phase, it is computationally consuming. We use a relatively lightweight network, which contains two groups of cells, for architecture search. Each group is composed of two normal cells and a reduction cell. In total, six cells are used for the search stage, which is illustrated at the top of Fig. 2.

Multiscale information is vital in deepfake forensics [19]. We try to preserve this valuable multiscale information for the final detection. Thus, we use a deeper network to gradually decode the feature maps. As shown in the bottom of Fig. 2, the detection network contains four groups of cells. Each group has two normal cells followed by a reduction cell. We adopt the cells obtained from the above search stage without any training weights. The detection network is trained on benchmark datasets for in-dataset and cross-dataset tests.

3.8. Discussion of the novelty

In this paper, we collect the forgery-oriented search space. The manipulation traces are too fragile to be discovered by ordinary layers in the neural networks. CDC can be used for face anti-spoofing, which indicates its effectiveness in extracting face features. However, previous methods only consider the fixed value of θ of the CDC. The value of θ controls the proportion of gradient information. The gradient information plays an important role in deepfake detection. Thus, we gradually change the value of θ in CDC in the search space. This

attempt encourages the search process to find the most valuable gradient information for deepfake detection. The detection results have been improved. Table 2 shows the performance comparison between our method and common search space, e.g., “AutoDFD [arXiv21]” [23]. Table 4 shows the performance improvements of our search space compared with “vanilla CNN operations”.

We also introduce a novel performance estimation strategy to select more general architectures. Some NAS methods are prone to overfitting, while generalization capability is of importance for facial manipulation detection. The distance between the error of the training set and the error of the test set is a good indicator of generalization ability for a classification task. This idea carries over to the search process of NAS. We introduce a novel performance estimation indicator in Eq. (5) to encourage the error in the performance estimation phase to be close to the error in the search phase. This attempt helps us to find a suitable architecture with the minimum errors in the performance estimation phase, which improves the performance for cross-dataset deepfake detection. The experimental results are shown in Table 5.

4. Experiment

In this section, we describe the experiments in detail. First, we establish the experimental setup, including datasets, evaluation metrics, compared methods, and implementation details. Then, we analyze the comparison with state-of-the-art approaches. After that, we give the ablation study by trying different choices of search space, search strategy, cell connection, and input size. Finally, we illustrate visualization analysis and discuss the limitations.

4.1. Experimental setup

4.1.1. Datasets

We evaluate our method on four publicly available datasets. FaceForensics++ (FF++) [41] consists of four forgery approaches, e.g., DeepFakes (DF), Face2Face (F2F), FaceSwap (FS), and NeuralTextures (NT). This dataset covers two compression rates, e.g., high-quality (C23) and low-quality (C40). Celeb-DF [42] contains 590 real and 5639 forged videos totally from YouTube website. These clips are generated by advanced synthesis techniques. WildDeepfake [43] collects 7314 real-world face sequences from the Internet. The duration of the video clip changes dramatically. DFDC-preview [44] is composed of more than 5000 videos with various manipulation methods. The faces in the video may be partially forged.

In-dataset search only uses FF++ for searching neural architectures. Cross-dataset search adopts four datasets alternately. For in-dataset scenes, after determining the network structure, the models are trained and tested on the above four datasets, respectively. For cross-dataset experiments, we train the networks on FF++ dataset and test on the other three datasets to verify the generalization ability. In FF++ dataset, following the official splits, we use 720 videos for training, 140 videos for validation, and 140 videos for test. Each video takes 20 frames for training and 10 frames for validation and test. For the other three datasets, we divided them into the training set, validation set, and test set according to the ratio of 8:1:1.

4.1.2. Evaluation metrics

As a common practice in this field [9], we apply two widely used metrics, accuracy score (Acc), and area under the receiver operating characteristic curve (AUC) for evaluation. Their definition can be formulated as follows,

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (10)$$

$$\text{AUC} = \int_{x=0}^1 \text{TPR}(\text{FPR}^{-1}(x))dx, \quad (11)$$

where TP, TN, FP, and FN denote the number of correctly predicted real frames, accurately predicted fake frames, wrongly predicted real frames, and mistakenly predicted fake frames, respectively, TPR and FPR represent true positive rate and false positive rate, respectively.

4.1.3. Compared methods

The proposed method is compared with 23 face forgery detection frameworks, including

- **Meso4** [45] focuses on the mesoscopic properties of images.
- **MesoInception4** [45] replaces the first two convolutional layers of Meso4 by a variant of the inception module [46].
- **RNN** [47] uses a CNN to extract frame-level features, and train a RNN that learns to classify a video.
- **FWA** [48] is inspired by the limited resolutions of the generated images of deepfake algorithm.
- **Xception** [41] is a face forensic method based on Xception-Net [49].
- **Multi-Task** [50] denotes the multi-task learning network including an encoder and a Y-shaped decoder.
- **Capsule** [51] equips with a capsule network to detect various kinds of spoofs.
- **EfficientNet-B4** [52] is a forensic pipeline, adopting EfficientNet [52] as backbones.
- **F3-Net** [53] is short for frequency in face forgery network, which employing two frequency-aware clues.
- **Multi-Attention** [54] framework consists of spatial attention heads, textural feature enhancement block, and multi-level feature aggregation.
- **LTW** [55] is based on the meta-weight learning algorithm, which configures different weights for face images.
- **FT-TS** [56] is composed of two-stream backbones, analyzing from frame-level and temporality-level of videos.
- **AutoDFD** [23] employs neural architecture search for deepfake detection.
- **Finfer** [57] is a frame inference-based detection method for high-visual-quality deepfake detection.
- **RECCE** [58] focuses on the common compact representations of genuine faces based on reconstruction-classification learning.
- **NoiseDF** [59] emphasizes the underlying forensic noise traces left behind the deepfake videos.
- **MRL** [60] formulates deepfake detection as a graph classification problem.
- **F2Trans** [21] develops a high-frequency fine-grained Transformer to explore the fine-grained forgery traces.
- **ISTVT** [22] adopts an interpretable spatial-temporal video Transformer for face forgery detection.
- **DDL-STC** [9] designs a dynamic fine-grained difference capture module and a multi-scale spatio-temporal aggregation module.
- **FedForgery** [61] uses variational autoencoder to formulate residual federated learning.
- **ForensicSymmetry** [10] introduces a dual-stream structure to determine whether two symmetrical face patches contain the same or different natural features.
- **SFIC** [6] constructs new backbone networks via space-frequency interactive convolution.

We directly report the quantitative results in the published papers if accessible.

4.1.4. Implementation details

The facial areas are extracted by the `face_recognition` toolbox. Due to GPU memory limitations, the input pictures are resized to 32×32 for the search phase. For in-dataset search, a half of the data in FF++ is used for searching architecture hyper-parameters, while the other half is used for updating network weights. For cross-dataset search, we randomly selects 2000 samples from each dataset to participate in the search in each epoch. The search process uses the Adam optimizer [62] for 65 epochs. The batch size is set to 96 for the search phase. The initial number of channels is 16, and the initial rate of channel sampling is 1/8. The running time of search stage is about 3 h

Table 1

Comparisons of the in-dataset results on Celeb-DF, WildDeepfake, and DFDC-preview datasets. The accuracy and AUC scores (%) are reported.

Method	Celeb-DF		WildDeepfake		DFDC-pre	
	Acc	AUC	Acc	AUC	Acc	AUC
Meso4 [45] [WIFS18]	67.53	66.17	64.47	66.5	75.39	76.47
RNN [47] [AVSS18]	71.20	86.52	66.87	67.35	75.02	77.48
FWA [48] [CVPRW19]	64.73	60.16	55.46	57.92	73.25	72.97
Xception [41] [ICCV19]	90.34	89.75	75.26	80.89	79.32	81.58
FT-TS [56] [TCSVT21]	80.74	86.67	68.78	68.09	63.85	64.03
Finfer [57] [AAAI22]	90.47	93.30	75.88	81.38	80.39	82.88
DFD-NAS (Ours _{in})	94.92	99.59	78.25	85.45	80.80	88.72
DFD-NAS (Ours _{cross})	94.34	99.75	78.55	86.41	79.57	88.70

Table 2

Comparisons of the in-dataset results on FF++, and the cross-dataset results on Celeb-DF datasets. These methods are trained on FF++. The AUC scores (%) are reported.

Method	FF++	Celeb-DF
	AUC	AUC
Meso4 [45] [WIFS18]	84.70	54.80
MesoInception4 [45] [WIFS18]	83.00	53.60
RNN [47] [AVSS18]	90.13	63.56
FWA [48] [CVPRW19]	80.10	56.90
Xception [41] [ICCV19]	99.70	65.30
Multi-Task [50] [BTAS19]	76.30	54.30
Capsule [51] [ICASSP19]	96.60	57.50
EfficientNet-B4 [52] [ICML19]	99.70	64.29
F3-Net [53] [ECCV20]	98.10	65.17
Multi-Attention [54] [CVPR21]	99.80	67.44
LTW [55] [AAAI21]	98.50	64.10
FT-TS [56] [TCSVT21]	92.47	65.56
AutoDFD [23] [arXiv21]	91.71	66.48
Finfer [57] [AAAI22]	95.67	70.60
RECCE [58] [CVPR22]	99.32	68.71
DDL-STC [9] [TIFS23]	-	71.36
ForensicSymmetry [10] [TIFS23]	99.44	58.10
SFIC-ResNet26 [6] [TIFS24]	96.38	70.70
DFD-NAS (Ours _{in})	98.04	67.62
DFD-NAS (Ours _{cross})	98.15	72.48

Table 3

Comparisons of the cross-dataset results on WildDeepfake, and DFDC-preview datasets. These methods are trained on FF++. The AUC scores (%) are reported.

Method	WildDeepfake	DFDC-pre
	AUC	AUC
Meso4 [45] [WIFS18]	59.74	59.30
RNN [47] [AVSS18]	67.03	59.37
FWA [48] [CVPRW19]	67.35	59.49
Xception [41] [ICCV19]	60.54	64.29
FT-TS [56] [TCSVT21]	59.82	59.09
Finfer [57] [AAAI22]	69.46	70.39
RECCE [58] [CVPR22]	64.31	69.06
NoiseDF [59] [AAAI23]	-	63.89
MRL [60] [TIFS23]	-	71.53
F2Trans-B (IL) [21] [TIFS23]	-	71.81
ISTVT [22] [TIFS23]	-	74.20
DDL-STC [9] [TIFS23]	-	73.08
FedForgery [61] [TIFS23]	74.59	-
ForensicSymmetry [10] [TIFS23]	-	54.62
SFIC-ResNet26 [6] [TIFS24]	-	63.85
DFD-NAS (Ours _{in})	74.32	74.85
DFD-NAS (Ours _{cross})	74.71	75.79

on an NVIDIA 3080Ti GPU due to channel sampling and pruning. After obtaining the determined architecture, we train the model on FF++ or the other three datasets for 150 epochs. It takes nearly 20 h for the training phase, when the input picture size is 64×64 and the batch size is 48. We adopt the SGD optimizer [63] with an initial learning rate of 0.025, a momentum of 0.9, a weight decay of $3e-4$.

Table 4

Ablation analysis of different search spaces. The accuracy and AUC scores (%) are reported.

Method	Celeb-DF	
	Acc	AUC
Vanilla CNN operations (PC-DARTS [39])	92.42	97.56
Fixed $\theta = 0.7$ in CDC	93.06	98.23
Ours _{cross}	94.34	99.75

Table 5

Ablation analysis of different search strategies. “IND” denotes the proposed performance estimation indicator. The accuracy and AUC scores (%) are reported.

Method	Pruning	Indicator	Search time (GPU-Days)	Celeb-DF	
				Acc	AUC
w/o pruning+IND	✗	✗	0.31	93.87	99.62
w/o IND	✓	✗	0.13	94.06	99.50
w/o pruning	✗	✓	0.33	94.12	99.65
Ours _{cross}	✓	✓	0.13	94.34	99.75

4.2. Comparison with state-of-the-art methods

“Ours_{cross}” denotes the performance using cross-dataset search strategy. “Ours_{in}” means that the search process only uses FF++ dataset without cross-dataset search strategy. After the search process, the architecture obtained by NAS is trained and tested on the benchmark datasets. When training and test using the same dataset, the results are shown in Table 1 and the first column of Table 2. When the architecture obtained by NAS is trained on the FF++ dataset and tested on the other datasets, the results are shown in the second column of Tables 2 and 3.

4.2.1. In-dataset evaluation

After the search process, the architecture is trained and tested on the above datasets, respectively. The results are shown in Table 1 and the first column of Table 2. Except for FF++, the proposed method outperforms other baselines in this scenario. “Ours_{in}” means that the search process only uses FF++, while “Ours_{cross}” denotes cross-dataset search. Note that our method also beats another NAS framework, e.g., AutoDFD [23].

4.2.2. Cross-dataset evaluation

To validate the generalization capability, we train the network on FF++. Then, we test this model on other datasets. As listed in the second column of Tables 2 and 3, the proposed method achieves competitive performance with cross-dataset search. Considering the strict definition of cross-dataset test, we denote the situation when the search stage is only conducted on FF++ as “Ours_{in}”, our method still shows its effectiveness.

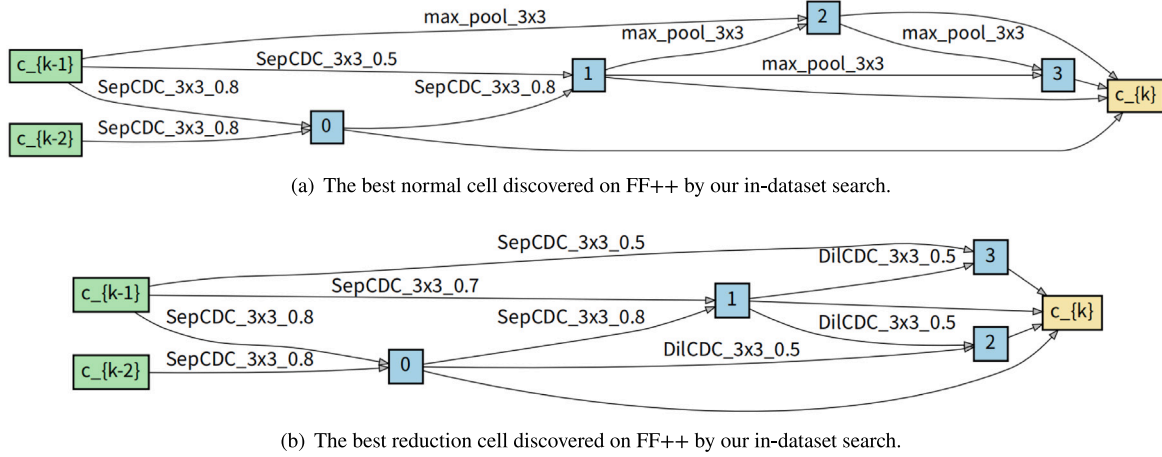


Fig. 5. Detailed structure of the best cells discovered on FF++ by our in-dataset search. (a) normal cell. (b) reduction cell. It is obvious that the operations are different from vanilla CNN. Diverse operations and θ values in the forgery-oriented search space are necessary.

Table 6

Ablation analysis of different cell connections. “N” represents a normal cell, and “R” denotes a reduction cell. The accuracy and AUC scores (%) are reported.

Connections between cells	Celeb-DF	
	Acc	AUC
$N \times 6 + R + N \times 6 + R + N \times 6$ (PC-DARTS [39])	93.55	97.78
$N \times 6 + R + N \times 6 + R$	93.77	98.33
$N \times 8 + R + N \times 8 + R$	94.05	98.68
$N \times 4 + R + N \times 4 + R + N \times 4 + R$	94.10	95.99
Ours _{cross}	94.34	99.75

4.3. Ablation study

In this subsection, we analyze the necessity of some key settings in our method, including search space, search strategy, cell connection, and input size. In the following ablation experiments, we only consider the performance of in-dataset evaluation. The final detection networks are trained on Celeb-DF [42].

4.3.1. Different search space

The forgery-oriented search space plays an important role in our method. As shown in Table 4, our search space is more effective than vanilla CNN in PC-DARTS [39]. Central difference has a positive influence apparently. Various θ values also perform better than the default value $\theta = 0.7$.

4.3.2. Different search strategy

We also compare the effects of the search strategy in Table 5. “IND” denotes the proposed performance estimation indicator. The proposed method adopts pruning and a novel performance estimation indicator, making the whole algorithm more effective for this forensics task. Concretely, pruning saves nearly half of the search time. The proposed new indicator also contributes to better performance.

4.3.3. Different cell connection

Since the multiscale information conveys vital clues for forgeries [19], we gradually downsample the resolutions between cells. In Table 6, “N” represents a normal cell, and “R” denotes a reduction cell. The number of reduction cells has an impact on performance. As the number of reduction cells increases, the results become better. The proposed C2PN adopts four reduction cells ($N \times 2 + R + N \times 2 + R + N \times 2 + R + N \times 2 + R$)

to progressively downsample feature maps. This choice obtains the best results among all the approaches, including the cell connection method in PC-DARTS [39].

4.3.4. Different input size

In this field, the input size of most methods is 256×256 . If the input size is 256×256 for training, the batch size can only be set to 4, which takes about 2.6 h to train an epoch. If using a 64×64 image for training, the batch size can be set to 48, which only takes about 8 min to train an epoch. When the input is 64×64 , the FLOPs of our model is 1.744G and the parameter quantity is 13.37M. Due to GPU memory limitations, we choose smaller images as inputs. Though our input image resolution is smaller than other baselines, we still achieved competitive performance.

4.4. Visualization analysis

4.4.1. Detailed structure of the best cells

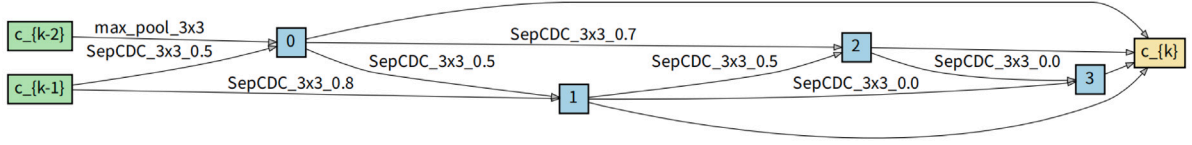
The best cells obtained by in-dataset search and cross-dataset search are shown in Figs. 5 and 6, respectively. The normal cell preserves the resolution, while the reduction cell downsamples the feature maps by half. The operations in both normal cells and reduction cells are obviously different from vanilla CNN, which indicates the significance of central difference convolution. Diverse operations and θ values are necessary. In Fig. 5, the reduction cell adopts dilate convolution to enhance the multiscale information of forged faces.

4.4.2. Detailed structure of the best cells searched by PC-DARTS

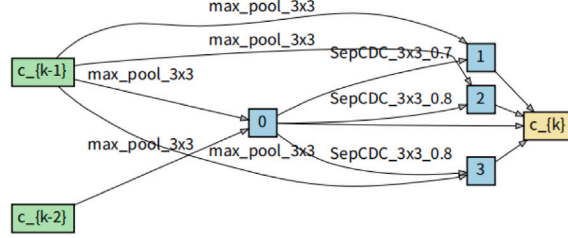
We use PC-DARTS [39] as a baseline for visualization analysis. The normal cell and the reduction cell searched by PC-DARTS are shown in Fig. 7. In these cells, the operations mainly consist of pooling and skip connections. The structures of these cells are obviously different from the cells produced by our method.

4.4.3. Visualization analysis on activation maps

The Grad-CAM [64] visualizations on FF++ datasets are illustrated in Fig. 8. These activation maps are generated from the outputs of the fourth reduction cell in the final detection networks. As we can see, most response areas are in the face regions, especially in the nose and the mouth. The clues of different forgery types locate in distinct regions. For example, the activations of Face2Face always cover the whole face, including mouth areas. The activations of FaceSwap

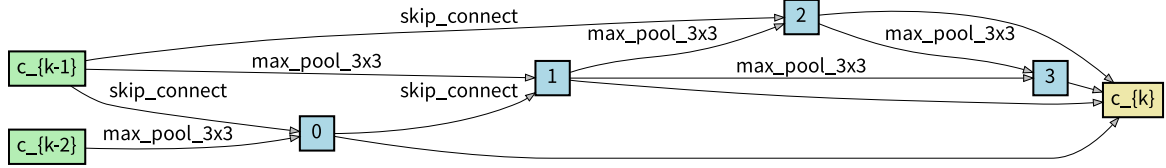


(a) The best normal cell discovered on FF++ by our cross-dataset search.

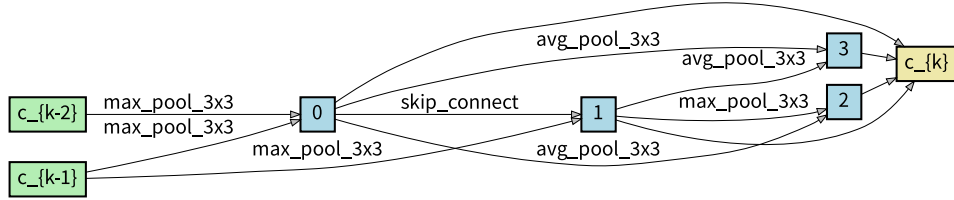


(b) The best reduction cell discovered on FF++ by our cross-dataset search.

Fig. 6. Detailed structure of the best cells discovered on FF++ by our cross-dataset search. (a) normal cell. (b) reduction cell. It is obvious that the operations are different from vanilla CNN. Diverse operations and θ values in the forgery-oriented search space are necessary.



(a) The best normal cell discovered by PC-DARTS.



(b) The best reduction cell discovered by PC-DARTS.

Fig. 7. Detailed structure of the best normal cell and reduction cell discovered by PC-DARTS [39]. It is obvious that the operations are different from the cells obtained by our algorithm.

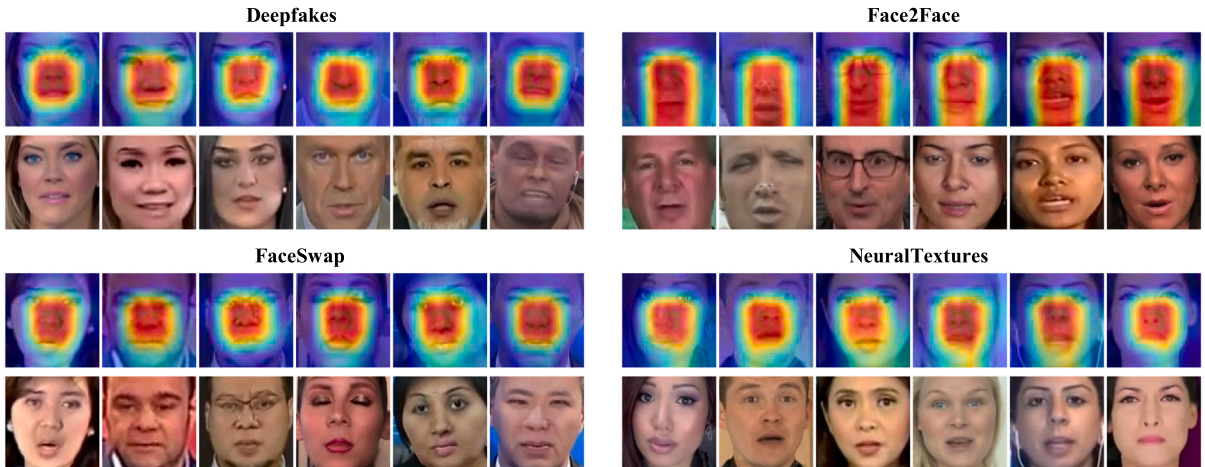


Fig. 8. The activation map visualization on FaceForensics++ dataset [41].

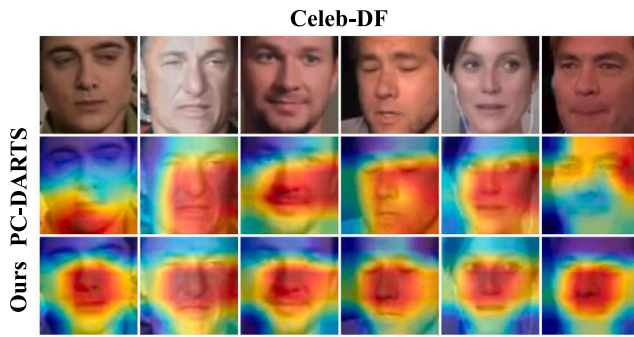


Fig. 9. The activation map visualization on Celeb-DF dataset [42].

are usually concentrated in the central facial regions. Note that the regions reflected by Grad-CAM are the basis for determining the final classification results, rather than exactly tampered regions.

The Grad-CAM [64] visualizations on Celeb-DF dataset [42] are illustrated in Fig. 9. The first row in Fig. 9 shows the fake faces in the videos. The second row denotes the activation maps when we replace the proposed search space with the search space of PC-DARTS. The last row in Fig. 9 illustrates the activation maps of the proposed method, which are generated from the outputs of the fourth reduction cell in the final detection networks.

As we can see, our method performs better. Compared with PC-DARTS, the response to the tampered areas is more accurate in the proposed method. These visualizations prove that the proposed forgery-oriented search space is more effective for facial manipulation detection.

4.5. Discussion of the limitation

The main objective of this article is to explore the potential of NAS for deepfake detection without manual intervention. Nonetheless, the automatically searched neural network is still based on the CNN architecture. Compared with graph-based frameworks and Transformer-based models, our method did not show significant performance improvement on some datasets.

However, our method utilizes fewer parameter quantities and achieves comparable results. The parameter quantity of our model is 13.37M.

SFDG [65] adopts EfficientNet-b4 (#Params: >19M) as backbone. Our method performs better in cross-dataset test on WildDeepfake dataset (AUC: 74.71 vs. 69.27) and DFDC dataset (AUC: 75.79 vs. 73.64). PEL [66] employs a dual-stream framework using EfficientNet-b4 as backbones (#Params: >38M). We obtain better performance in cross-dataset circumstances on Celeb-DF (AUC: 72.48 vs. 69.18), WildDeepfake (AUC: 74.71 vs. 67.39), DFDC (AUC: 75.79 vs. 63.31). Compared with UIA-ViT [67], we obtain comparable results in DFDC-pre dataset (AUC: 75.79 vs. 75.80) in cross-dataset setting. The parameter quantity of our network is much fewer than the ViT-Base structure in UIA-ViT (#Params: 13.37M vs. 86M).

In practical application scenarios on mobile devices, methods with smaller parameter quantities are more convenient to deploy and have better practicality.

5. Conclusion

In this paper, we propose an end-to-end face forgery detection framework based on NAS, which can automatically design suitable neural networks for this task. First, we create a forgery-oriented search space to find appropriate operations for deepfake detection. Then, we take the actual estimation error as guidance to improve the search generalizability. Besides ordinary search manners, we also consider cross-dataset search to obtain a more general neural network architecture.

Eventually, we connect the cells obtained by NAS with skip connections in a cascaded pyramid way. Extensive experimental results demonstrate the proposed method achieves competitive performance compared with state-of-the-art approaches, especially good generalization ability in cross-dataset settings.

In the future, since the Transformer architecture [68] has shown significant performance improvements in many fields, we will try to further expand the search space to make DFD-NAS compatible with Transformer modules and thus search for more promising architectures. Furthermore, considering the convergence can indicate the superiority of the searched architectures in an unsupervised NAS setting, we would like to extend our NAS framework in the unsupervised learning manner [69].

CRedit authorship contribution statement

Xiao Jin: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Wen Yu:** Software, Data curation. **Dai-Wei Chen:** Data curation, Software. **Wei Shi:** Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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